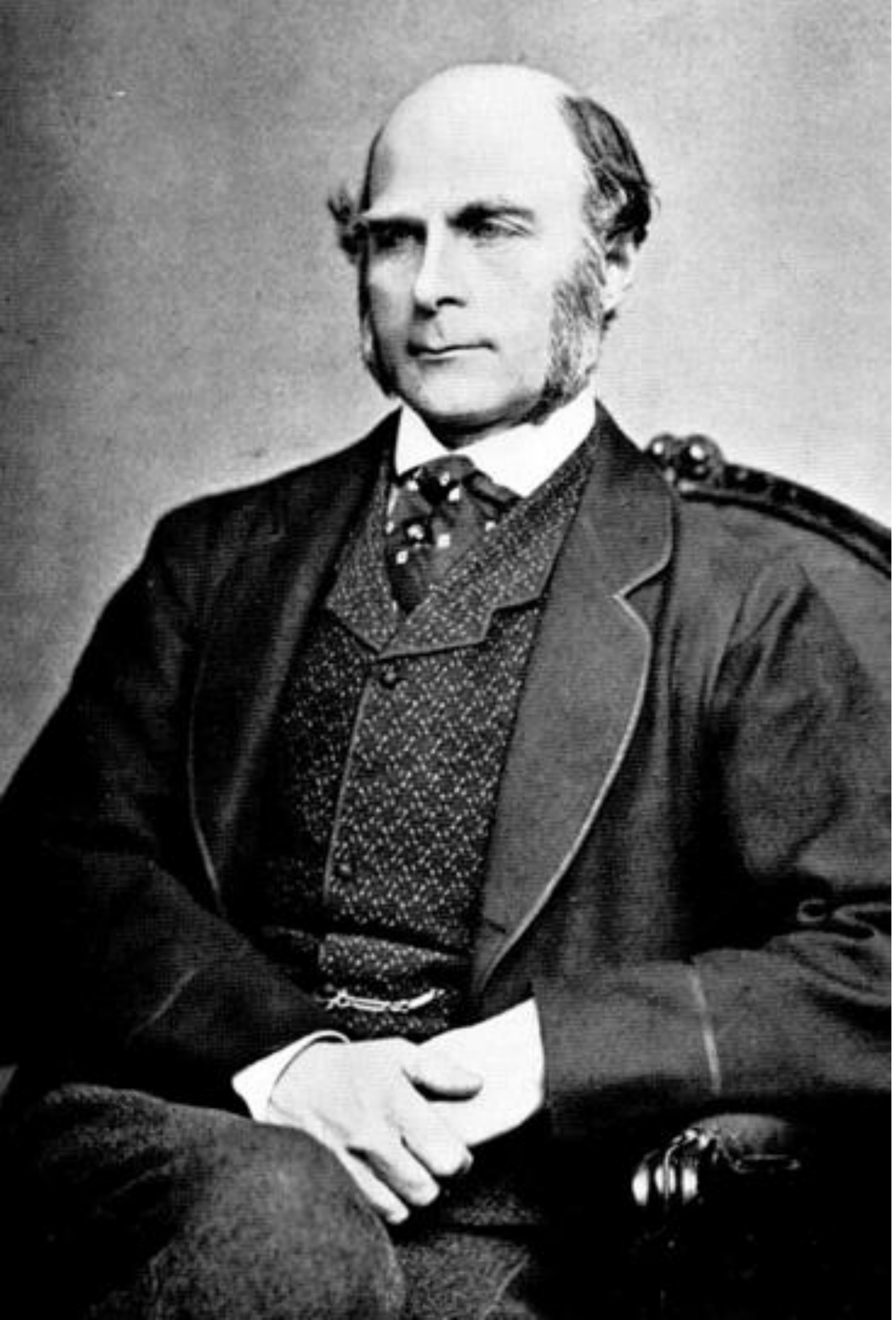
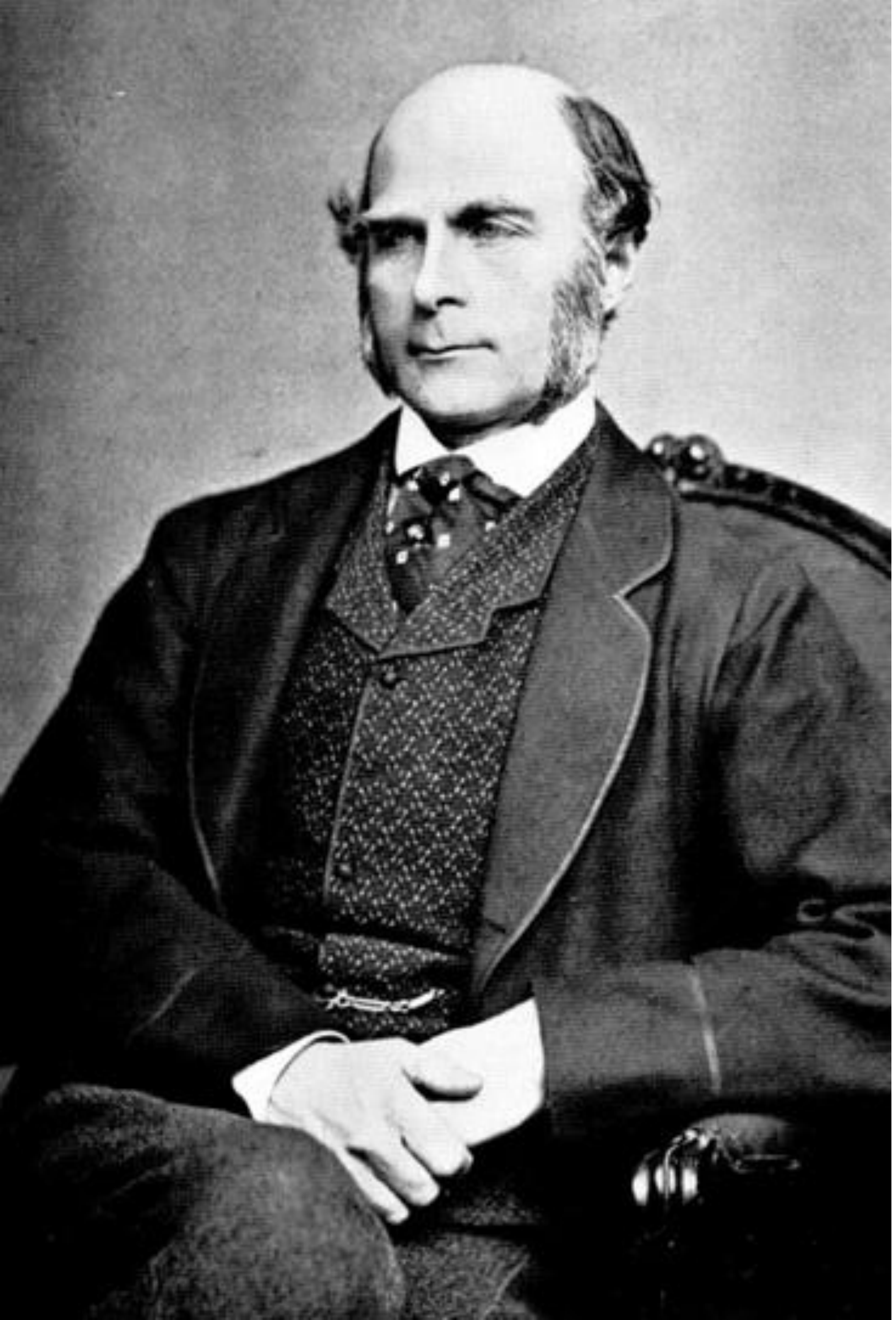
Regression Analysis



History

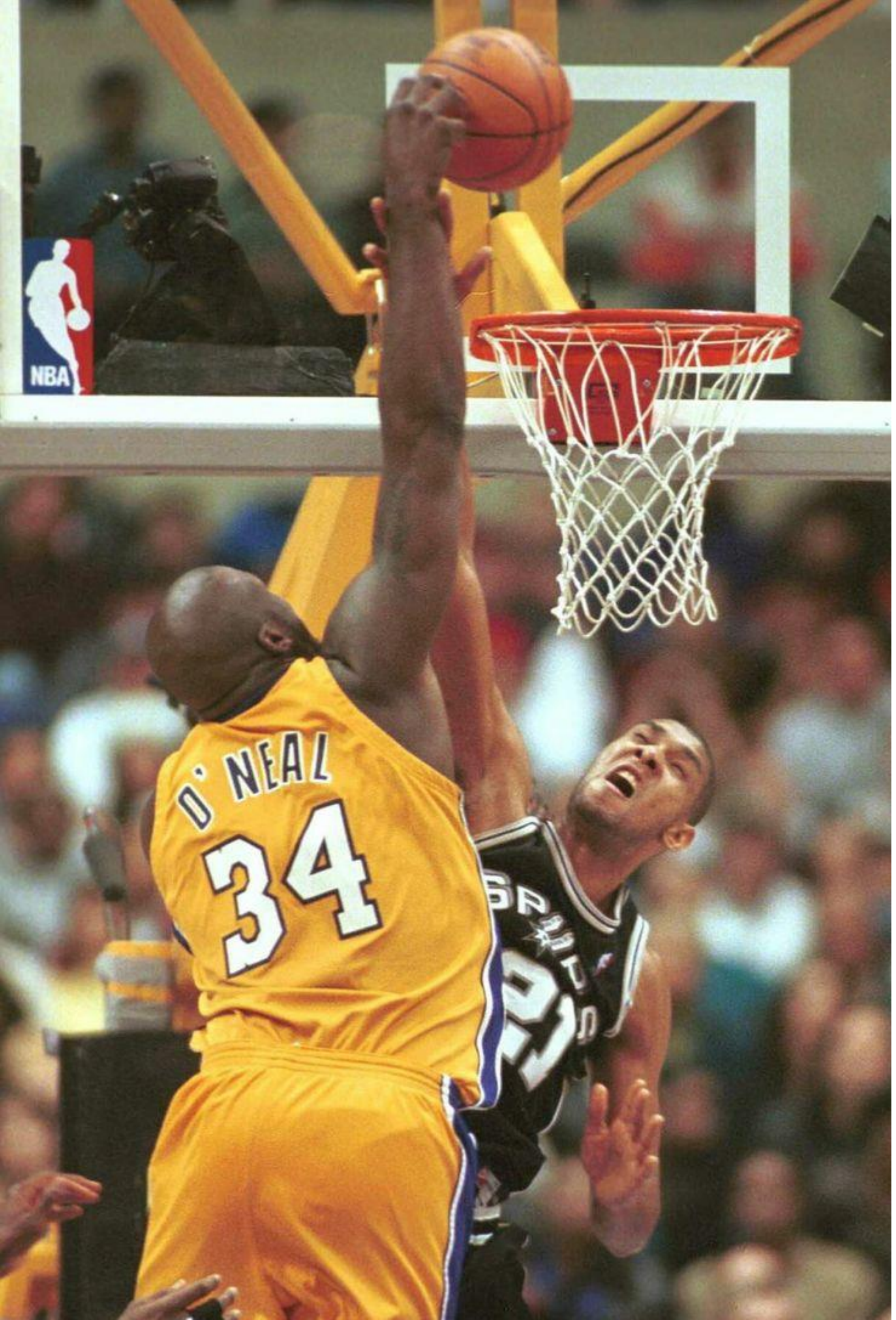
This all started in the 1800s with a guy named Francis Galton. Galton was studying the relationship between parents and their children. In particular, he investigated the relationship between the heights of fathers and their sons.



History

What he discovered was that a man's son tended to be roughly as tall as his father.

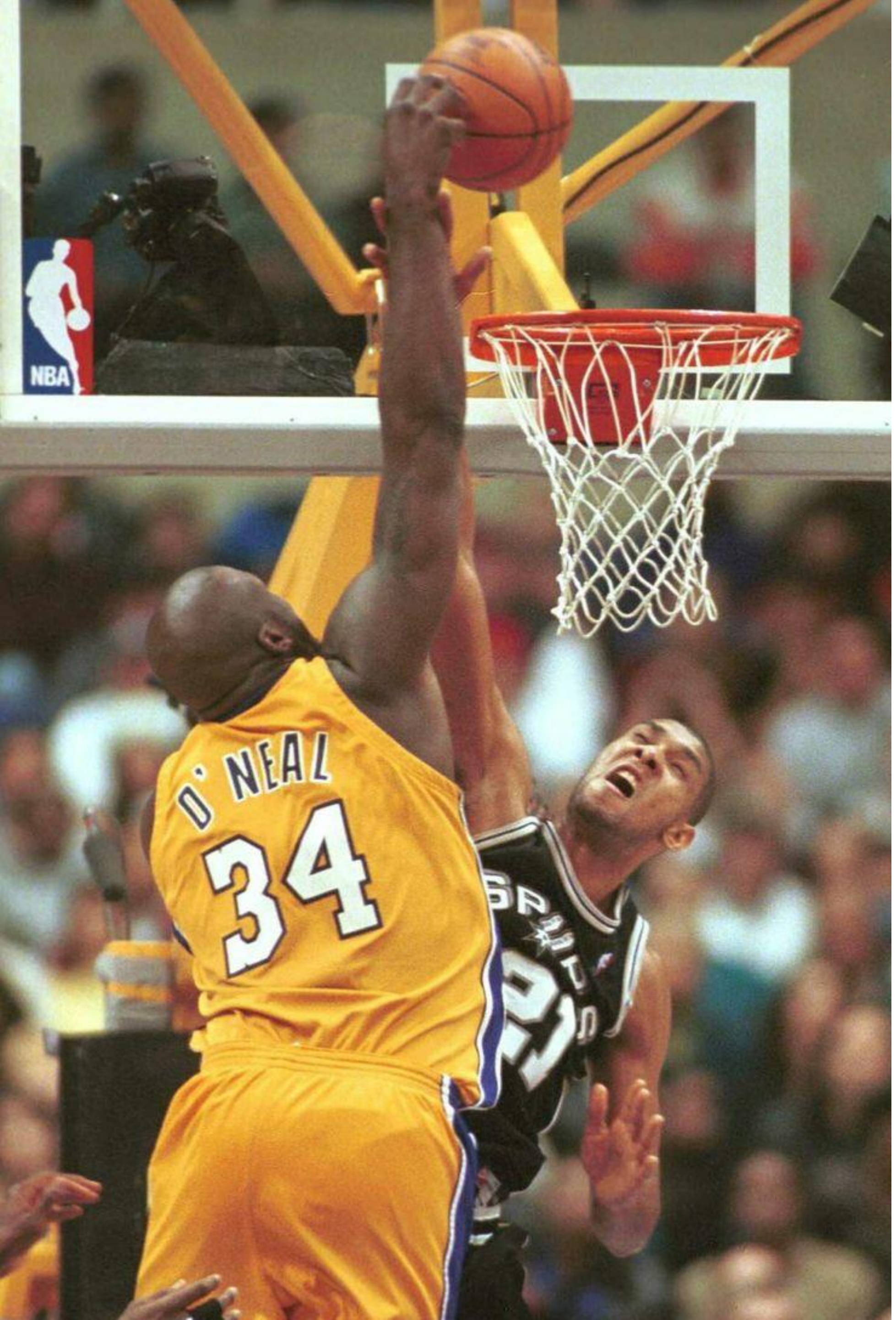
However Galton's breakthrough was that the son's height **tended to be closer to the overall average** height of all people.



Example

Let's take Shaquille O'Neal as an example. Shaq is really tall:7ft 1in (2.2 meters).

If Shaq has a son, chances are he'll be pretty tall too. However, Shaq is such an anomaly that there is also a very good chance that his son will be **not be as tall as Shaq**.



Example

Turns out this is the case: Shaq's son is pretty tall (6 ft 7 in), but not nearly as tall as his dad.

Galton called this phenomenon **regression**, as in "A father's son's height tends to regress (or drift towards) the mean (average) height."

Regression model

- Relation between variables where changes in some variables may “explain” or possibly “cause” changes in other variables.
- Explanatory variables are termed the **independent** variables and the variables to be explained are termed the **dependent** variables.
- Regression model estimates the nature of the relationship between the independent and dependent variables.
  - Change in dependent variables that results from changes in independent variables, ie. size of the relationship.
  - Strength of the relationship.
  - Statistical significance of the relationship.

Examples

- Dependent variable is retail price of gasoline in Regina – independent variable is the price of crude oil.
- Dependent variable is employment income – independent variables might be hours of work, education, occupation, sex, age, region, years of experience, unionization status, etc.

- Price of a product and quantity produced or sold:
  - Quantity sold affected by price. Dependent variable is quantity of product sold – independent variable is price.
  - Price affected by quantity offered for sale. Dependent variable is price – independent variable is quantity sold.

Recall: Covariance

$$\mathrm{cov}(x, y) = \frac{\sum_{i=1}^{n}(x_i - \bar{X})(y_i - \bar{Y})}{n-1}$$

Interpreting Covariance

$\text{cov}(X,Y) > 0 \rightarrow$ X and Y are positively correlated

$\text{cov}(X,Y) < 0 \rightarrow$ X and Y are inversely correlated

$\text{cov}(X,Y) = 0 \rightarrow$ X and Y are independent

Correlation coefficient

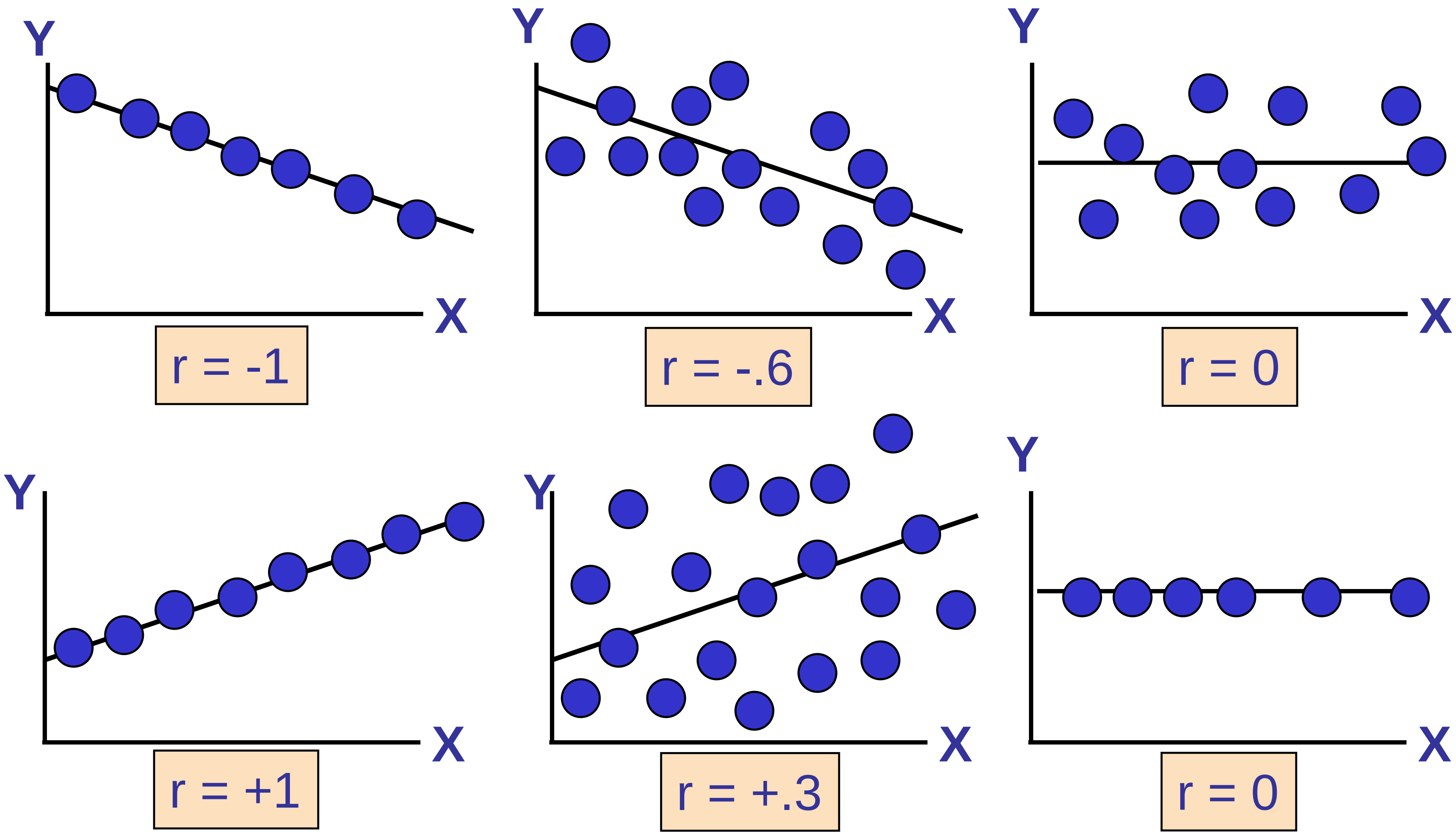
- Pearson's Correlation Coefficient is standardized covariance (unitless):

$$r = \frac{covariance(x,y)}{\sqrt{\operatorname{var} x}\sqrt{\operatorname{var} y}}$$

Correlation

- Measures the relative strength of the *linear* relationship between two variables
- Unit-less
- Ranges between –1 and 1
- The closer to –1, the stronger the negative linear relationship
- The closer to 1, the stronger the positive linear relationship
- The closer to 0, the weaker any positive linear relationship

Scatter Plots of Data with Various Correlation Coefficients



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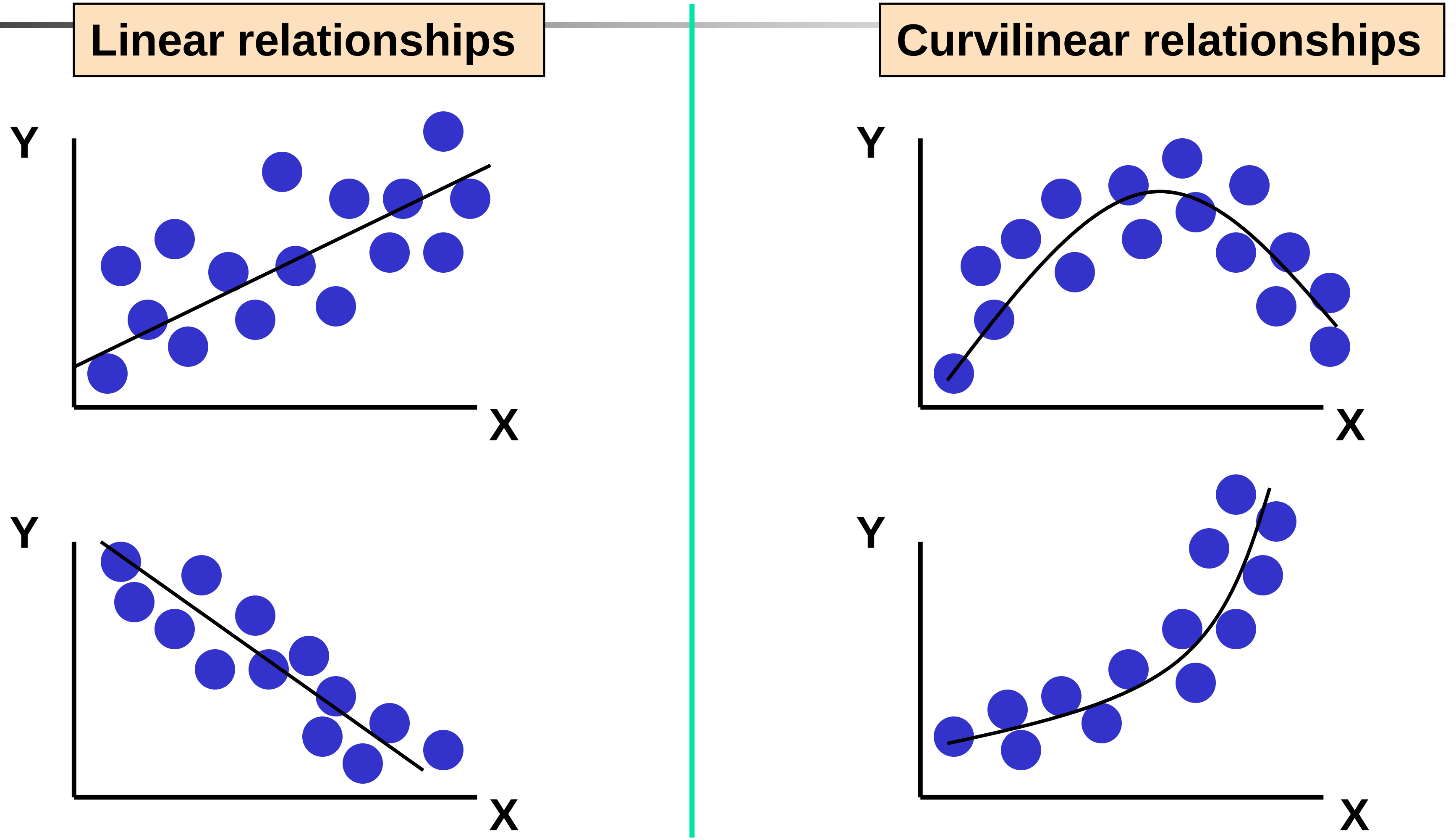
Linear Correlation



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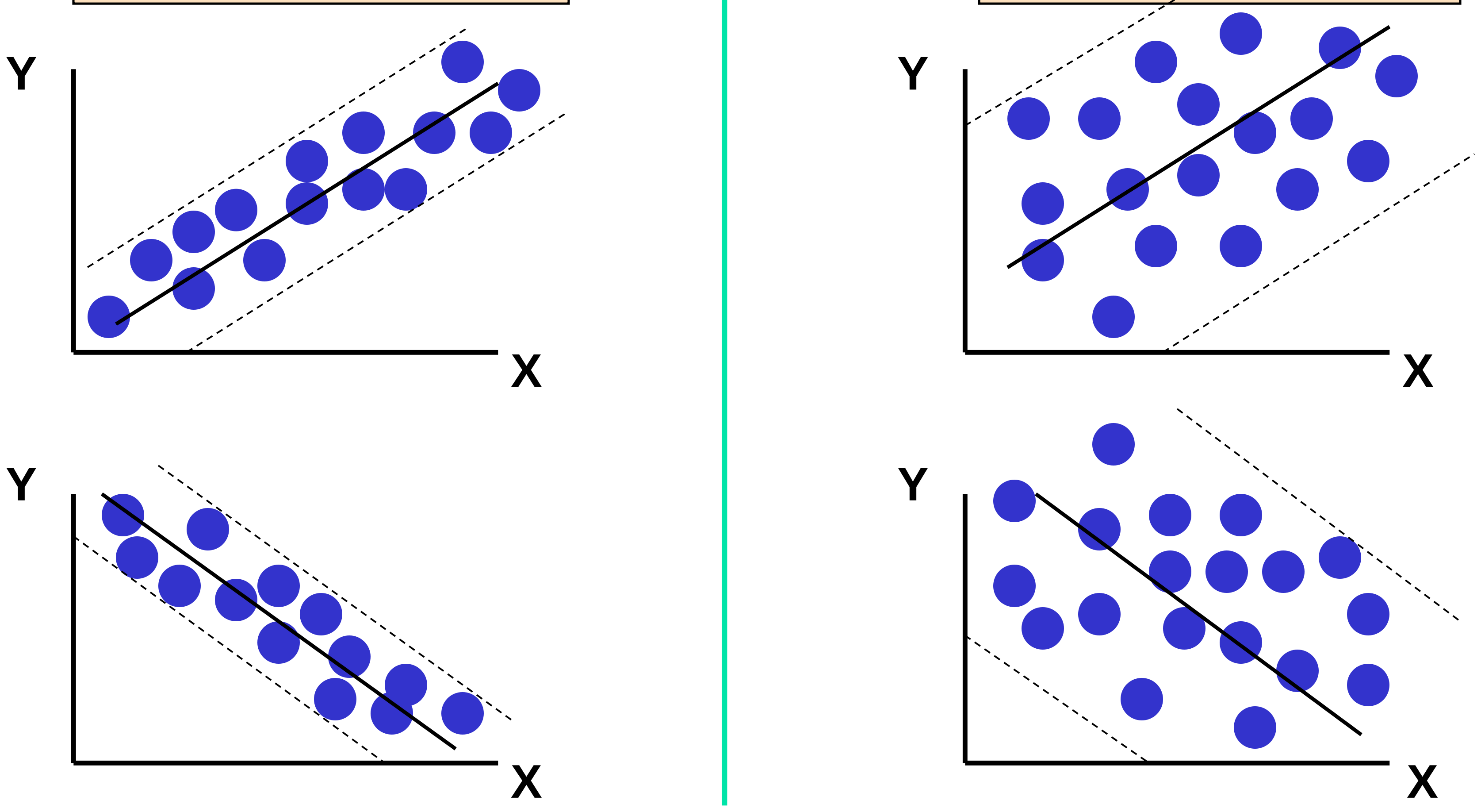
Linear Correlation

Strong relationships | **Weak relationships**



Calculating by hand…

$$\hat{r} = \frac{\mathrm{cov}ariance(x,y)}{\sqrt{\mathrm{var}\ x}\sqrt{\mathrm{var}\ y}} = \frac{\dfrac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n-1}}{\sqrt{\dfrac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}\sqrt{\dfrac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1}}}$$

Simpler calculation formula…

$$\hat{r} = \frac{\dfrac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\cancel{n-1}}}{\sqrt{\dfrac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{\cancel{n-1}}}\sqrt{\dfrac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{\cancel{n-1}}}} =$$

$$\frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}$$

$$\hat{r} = \frac{SS_{xy}}{\sqrt{SS_x SS_y}}$$

Numerator of covariance

Numerators of variance

Distribution of the correlation coefficient:

$$SE(\hat{r}) = \sqrt{\frac{1 - r^2}{n - 2}}$$

The sample correlation coefficient follows a T-distribution with n-2 degrees of freedom (since you have to estimate the standard error).

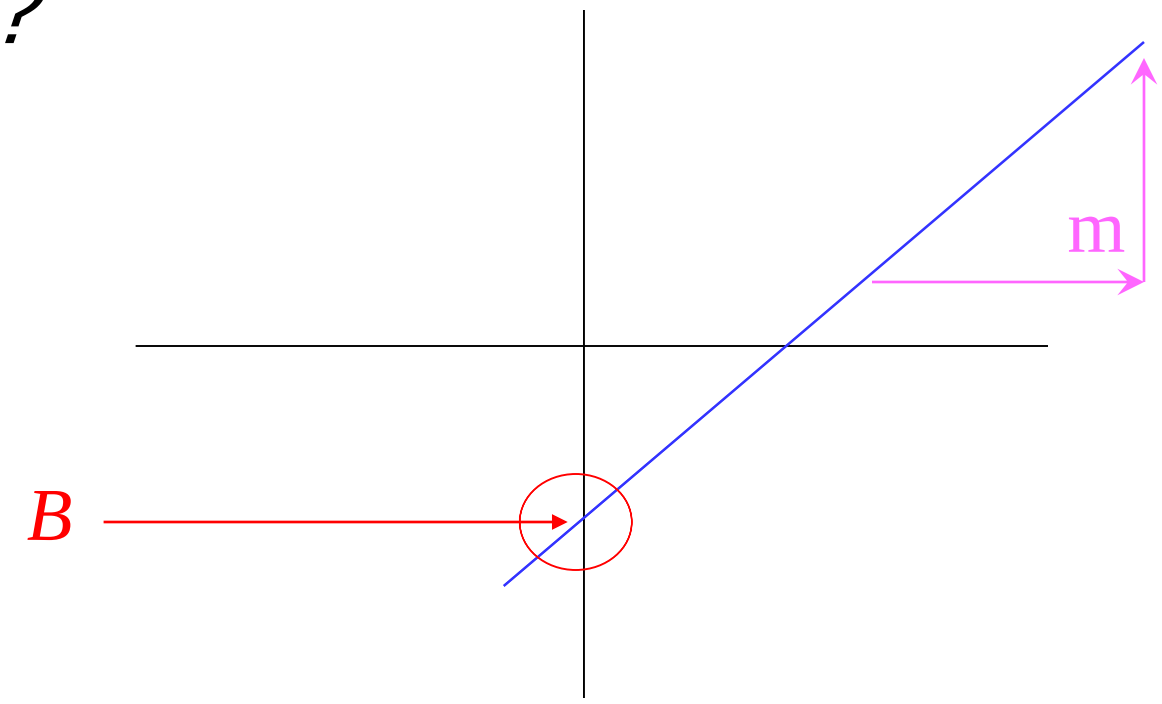
*note, like a proportion, the variance of the correlation coefficient depends on the correlation coefficient itself→substitute in estimated r

Linear regression

In correlation, the two variables are treated as equals. In regression, one variable is considered independent (=predictor) variable (*X*) and the other the dependent (=outcome) variable *Y*.

What is “Linear”?

- Remember this:
- *Y=mX+B?*



What's Slope?

A slope of 2 means that every 1-unit change in X yields a 2-unit change in Y.

Prediction

- If you know something about X, this knowledge helps you predict something about Y. (Sound familiar?…sound like conditional probabilities?)

Regression equation…

Expected value of y at a given level of *x*=

$$E(y_i / x_i) = \alpha + \beta x_i$$

Predicted value for an individual…

$$y_i = \hat{\alpha} + \beta * x_i \quad + \quad \text{random } \boxed{\text{error}_i}$$

Fixed – exactly on the line

Follows a normal distribution

Assumptions (or the fine print)

- Linear regression assumes that…
  - 1. The relationship between X and Y is linear
  - 2. Y is distributed normally at each value of X
  - 3. The variance of Y at every value of X is the same (homogeneity of variances)
  - 4. The observations are independent

Regression Analysis (Formal Definition)

- A statistical methodology
- Used to model the relationship between one or more **independent (predictor)** variable and a **dependent (response)** variable
  - **Predictor variables**: the attributes describing a tuple
  - **Response variable**: what we want to predict
- Many prediction problems can be solved using **linear regression**
- A **non-linear** problem can be converted a linear one

Linear Regression

- **Straight-line regression** analysis involves
  - A single predictor variable
  - A response variable

$$y = b + wx$$

  - The **variance** of y is **constant**
  - ***b*** and ***w*** are **regression coefficients**
    - *b*: Y-intercept
    - *w*: the slope of the line

- Regression coefficients can also be considered as weights

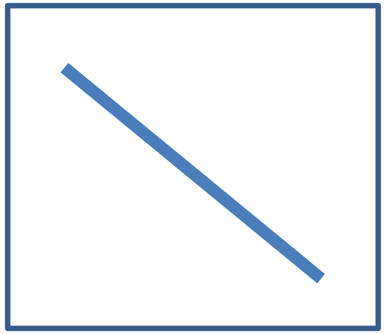
$$y = \beta_0 + \beta_1 x$$

- Need of estimating the regression coefficients

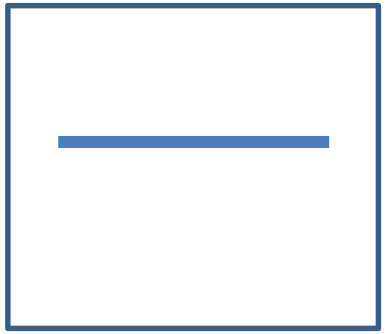
Notation

$\hat{y}$ is the predicted *y* value on the regression line

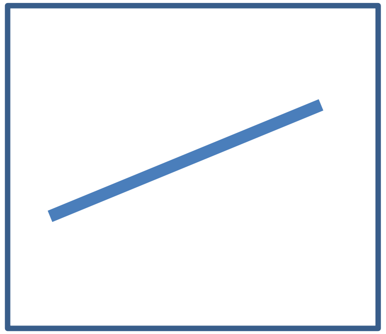
$$\boxed{\hat{y} = \text{intercept} + \text{slope } x} \qquad \boxed{\hat{y} = \beta_0 + \beta_1 x}$$



slope < 0



slope = 0



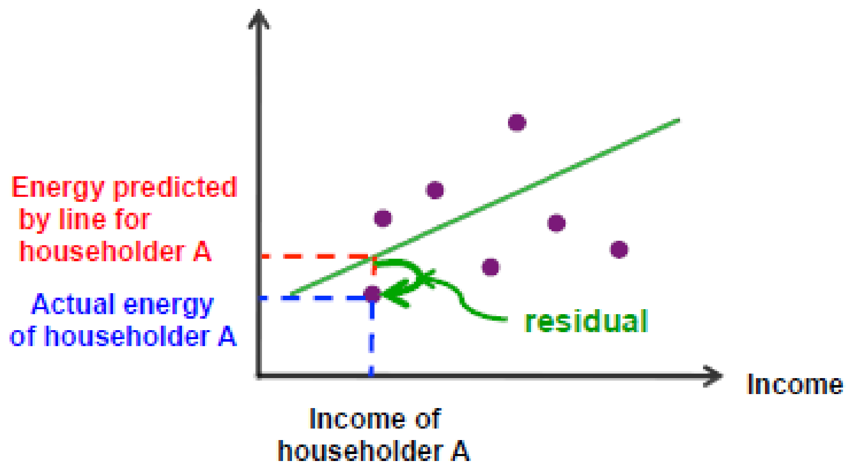
slope > 0

Not all calculators/software use this convention. Other notations include:

$$\hat{y} = \beta_0 + \beta_1 x$$

$$\hat{y} = \text{variable\_name } x + \text{constant}$$

Definition of Residuals



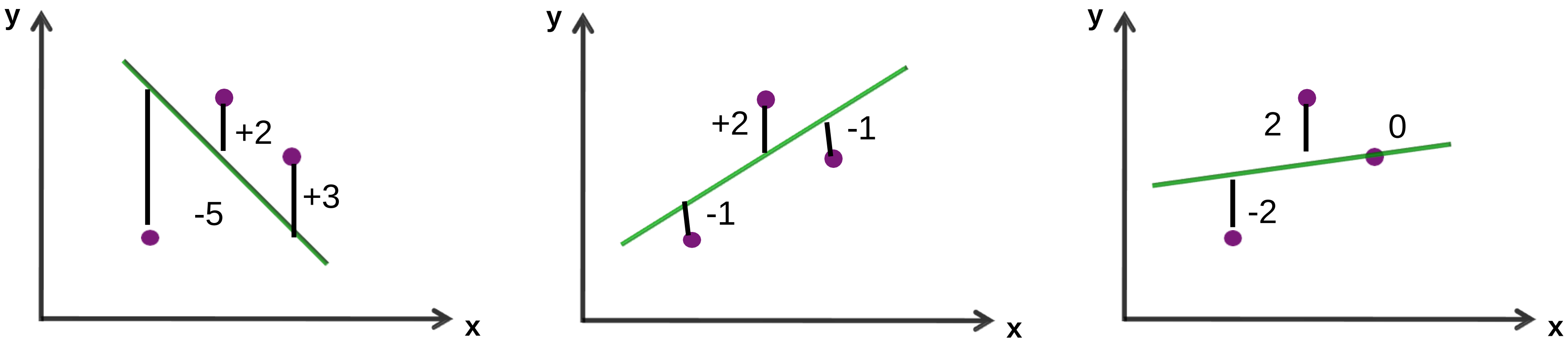
- Line: $\hat{E}_n = \alpha + \beta I_n$

  I denotes income
  n denotes household
  n=1,...,N

- Residual: $r_n = E_n - \hat{E}_n$

- For a good fit, we want residuals to be small

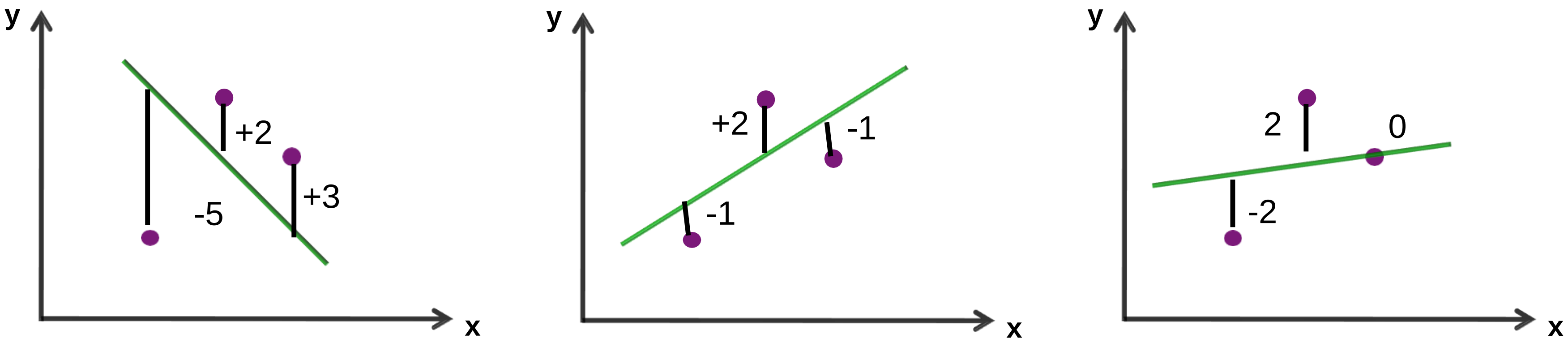
Which Residuals are smaller?



- Candidate criteria
  - The best line gives the **smallest sum of residuals**
  - Problem: Positives and negatives cancel out
    - -5+2+3=0
    - -1+2-1=0
    - -2+2+0=0
  - All lines are equally good by this criterion

$$\min\left|\sum_{n=1}^{N}(E_n-\hat{E}_n)\right|$$

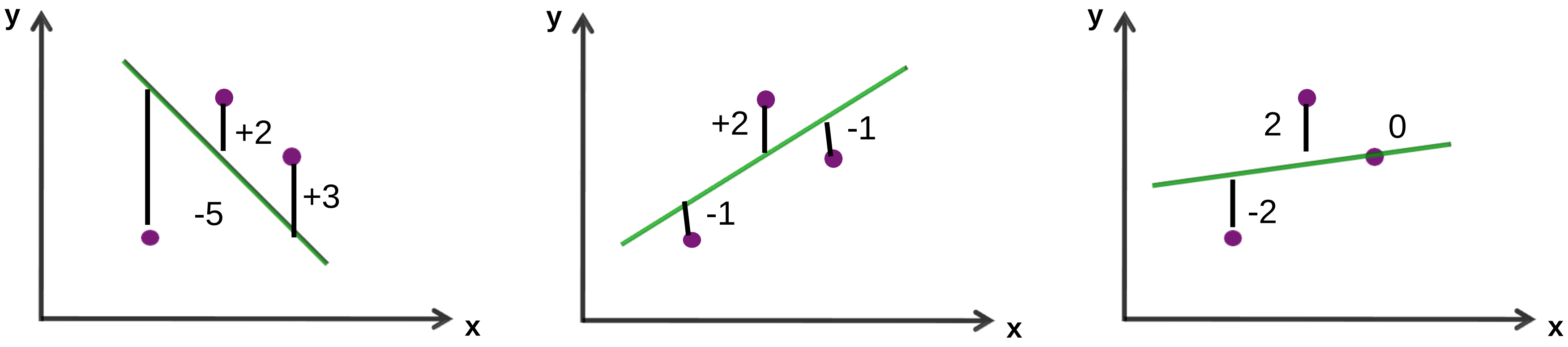
Which Residuals are smaller?



- Candidate criteria
  - The best line gives the **smallest sum of absolute deviations**
  - Problem: cannot always distinguish lines
    - 5+2+3=10
    - 1+2+1=4
    - 2+2+0=4
  - Different lines can be equally good by this criterion

$$\min \sum_{n=1}^{N} \left| E_n - \hat{E}_n \right|$$

Which Residuals are smaller?



- Candidate criteria
  - The best line gives **the smallest sum of squared residuals**
    - 25+4+9=38
    - 1+4+1=6
    - 4+4+0=8
  - weights outliers more, unbiased and efficient

$$\min \sum_{n=1}^{N} \left(E_n - \hat{E}_n \right)^2$$

Method of Least Squares

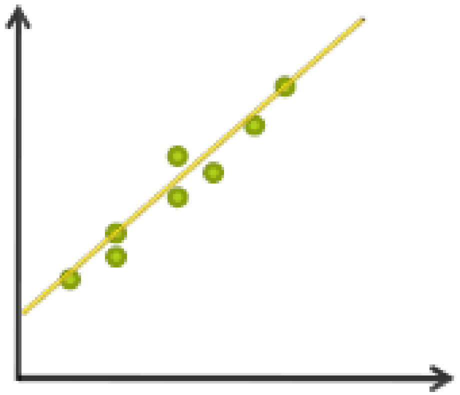
- Estimate the best-fitting straight line as the one that minimizes the error between the actual data and the estimate of the line

- Used to solve overdetermined systems (more equations than unknowns)
  - **f is the model function** where

$$y_i = f(x,\beta) = \beta_0 + \beta_1 x_i$$

  - Minimize the sum, S, of squared **residuals**

$$S = \sum_{i=1}^{|D|} r_i^{\,2} \qquad r_i = y_i - f(x_i, \hat{\beta})$$



- D: a set of training tuples with 1 predictor and 1 response each
  - (x_1, y_1)
  - (x_2, y_2)
  - …

Least Squares Estimation of β_0, β_1

- $\beta_0 \equiv$ Mean response when x=0 (y-intercept)
- $\beta_1 \equiv$ Change in mean response when x increases by 1 unit (slope)
- β_0, β_1 are unknown parameters (like μ)
- $\beta_0+\beta_1 x \equiv$ Mean response when explanatory variable takes on the value x
- Goal: Choose values (estimates) that minimize the sum of squared errors (SSE) of observed values to the straight-line:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x \qquad SSE = \sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2 = \sum_{i=1}^{n}\left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)\right)^2$$

Method of Least Squares

- The **minimum** of the sum of squares is found by setting the **gradient** to zero. If the model contains *m* parameters there are *m* gradient equations

$$\frac{\partial S}{\partial \beta_j} = 2\sum_i \frac{\partial r_i}{\partial \beta_j} = 0,\ j = 1,\ldots,m$$

- When m=2, the regression **coefficients** are **estimated** by:

$$\beta_1 = \frac{\sum_{i=1}^{|D|}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{|D|}(x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

where $\bar{x}$ is the man value of $x_1, x_2, \ldots, y_{|D|}$

$\bar{y}$ is the mean value of $y_1, y_2, \ldots, y_{|D|}$

Multiple Linear Regression

- Involve more than one predictor variables
- Model a response variable as linear function of **n** predictor variables **$A_1, A_2, \ldots, A_n$**
- D: a set of training tuples with **n predictors** and **1 response** each
  - (X_1, y_1)
  - (X_2, y_2)
  - …
  - $(X_{|D|}, y_{|D|})$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$$

- The method of least square is used to estimate the coefficients. However the computation becomes long
  - Use statistical software packages (e.g., SAS, SPSS, and S-Plus)

Multiple Linear Regression

- How to model data that does not show a linear dependence?
- Example: **polynomial regression**
  - Add polynomial terms to the basic linear model
  - Apply transformations to variables
  - Covert the nonlinear model to a linear one
- Consider a cubic polynomial relationship given by:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$$

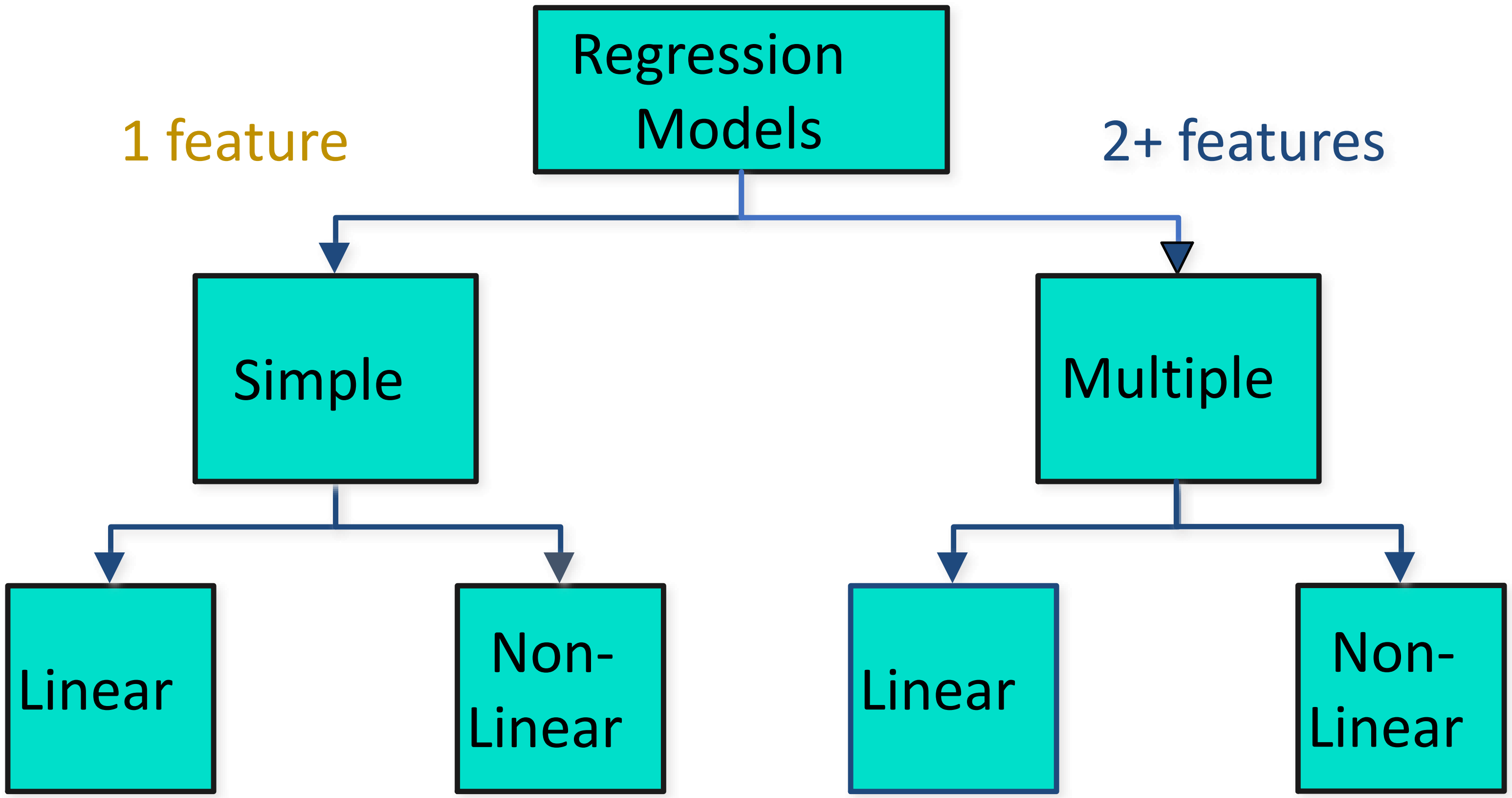
- To convert this equation to linear from, we define new variables

$$x_1 = x \qquad x_2 = x^2 \qquad x_3 = x^3$$

- The equation becomes

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

Types of Regression Models



Generalized Linear Models

- Represent the theoretical foundation on which the linear regression can be applied to model classification

- The variance of the response variable, is a function of the mean value of y, unlike the linear regression where the variance of y is constant

- Common types of generalized linear models include
  - Poisson regression
  - Logistic regression

Where Things Started?

- Binary classification guesses a binary output (**yes** or **no**) from input variables

- However, simply guessing “yes” or “no” is pretty crude — especially if there is no perfect rule

- Something which takes noise into account, and does not just give a binary answer, will often be useful

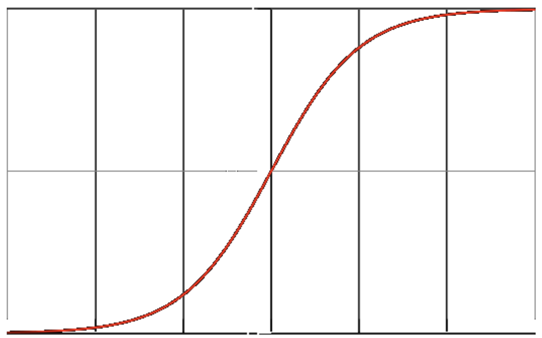
- In short, we want probabilities — which means we need to fit a stochastic model

Modeling Conditional Probabilities

- Need to have conditional distribution of the response Y, given the input variables, $Pr(Y \mid X)$
- This would tell us about how precise our predictions are
  - If our model says that there is a 51% chance of snow and it does not snow, that is better than if it had said there was a 99% chance of snow (though even a 99% chance is not a sure thing)
- We have seen how to estimate conditional probabilities non-parametrically (e.g., naive Bayes, Bayes Networks)
- We need to come up with a model for the joint distribution of outputs Y and inputs X, which can be quite time-consuming

Logistic Regression

- The logistic regression is used for binomial regression



- It predicts the probability of occurrence of an event by fitting data to a logistic curve

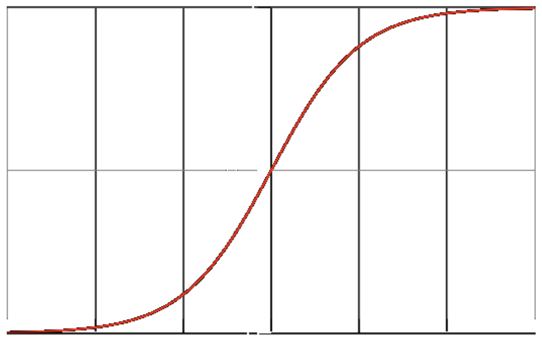
$$f(x) = \frac{e^x}{1 + e^x}$$

- x represents the exposure to some set of risk factors

- f(x) represents the probability of a particular outcome, given that set of risk factors.

Logistic Regression

- The variable x is a measure of the total contribution of all the risk factors (independent variables) used in the model and is known as the **logit**



$$f(x) = \frac{e^x}{1 + e^x}$$

- x is usually defined as

$$x = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k$$

- The logistic regression model is given by

$$P(Y = 1 | x_1, x_2, \ldots x_k) = \frac{e^{\beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k}}{1 + e^{\beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k}}$$

Likelihood Function for Logistic Regression

- Estimate parameters using Maximum Likelihood Estimator
  - **Data**: $y_j, x_{1j}, x_{2j}, \ldots, x_{pj}$, $j=1,2,\ldots,n$
  - Likelihood Function is given by:

$$L(\beta) = \prod_{j=1}^{n} \frac{e^{\beta_0 + \beta_1 x_{1j} + \ldots + \beta_p x_{pj}}}{1 + e^{\beta_0 + \beta_1 x_{1j} + \ldots + \beta_p x_{pj}}}$$

  - To simply the computation, we can maximize the log likelihood function
- To estimate the parameters
  - Compute the partial derivatives of the loglikelihood
  - Equate each partial derivative to zero, and solve the resulting nonlinear equations

Summary

- Numeric Prediction is the task of predicting **continuous values**
- **Regression analysis** is mostly used for prediction
- Regression can be of different forms **Linear** and **nonlinear**
- **Logistic regression** is used to model **binomial** regression